ERDŐS'S CONJECTURE ON $\sum f(n)/n$ IS FALSE: A 2-ADIC SHADOW OF OKADA'S CRITERION AND THE LEAST COUNTEREXAMPLE $q = 105^2$

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ABSTRACT. Let f be periodic modulo q with $f(n) = \pm 1$ for $q \nmid n$ and $f(n) = 0$ for $q \mid n$. Erdős conjectured, in a letter to Livingston, that $\sum_{n\geq 1} f(n)/n \neq 0$ whenever the series converges. By the theorem of Baker, Birch and Wirsing such a sum is either 0 or transcendental, so the conjecture is a transcendence statement; it had been proved for prime q, for $2\varphi(q) + 1 > q$ (Okada), for $q \equiv 3 \pmod 4$ (Murty–Saradha), and for a set of $q \equiv 1 \pmod 4$ of lower density at least 0.82 (Chatterjee–Murty). We show that the conjecture is false. For $q = 11025 = 3^2 5^2 7^2$ we give an explicit f, built from Legendre symbols on the divisor layers of $\mathbb{Z}/q\mathbb{Z}$, with $\sum f(n)/n = 0$; the proof reduces to eleven rational identities that can be checked by hand. Further counterexamples exist for $q = 165^2$, 315^2, 495^2 and 735^2, and the set of moduli carrying counterexamples is closed under multiplication.

The counterexamples come from a 2-adic principle. Since $\pm 1 \equiv 1 \pmod 2$, an Erdős function with vanishing sum is an *odd vector* of the Okada lattice, and we show that the existence of odd vectors is decided after a Galois descent to the maximal 2-power quotient of $(\mathbb{Z}/q\mathbb{Z})^\times$ (the "2-shadow"). Every odd square carries odd vectors. For every odd non-square $q \leq 13105$ there are none (exact computation with explicit certificates); this proves Erdős's conjecture, and hence the transcendence of all the sums, for these q, including 74 moduli not covered by earlier criteria. Consequently, for $q \leq 13105$ the conjecture fails exactly when $q = 11025$. We conjecture that the 2-adic obstruction holds for every odd non-square, which would confine all counterexamples to square moduli.

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1. Introduction

1.1. **The conjecture.** For an integer $q \geq 2$ call a function $f\colon \mathbb{Z} \to \{-1, 0, 1\}$ an *Erdős function modulo* q if f is q-periodic, $f(n) = \pm 1$ for $q \nmid n$ and $f(n) = 0$ for $q \mid n$. Put

$$L(1, f) = \sum_{n \geq 1} \frac{f(n)}{n}.$$

The series converges if and only if $\sum_{n=1}^{q} f(n) = 0$; since this is a sum of $q-1$ terms ± 1, convergence forces q to be odd.

Conjecture 1.1 (Erdős; see Livingston [6])**.** *If f is an Erdős function modulo q and $\sum_{n=1}^{q} f(n) = 0$, then $L(1, f) \neq 0$.*

The problem belongs to the circle of questions raised by Chowla [5] on the non-vanishing of $L(1, f)$ for periodic f. Baker, Birch and Wirsing [2] answered Chowla's question for prime moduli, which settles Conjecture 1.1 for prime q. For algebraic-valued periodic f the number $L(1, f)$ is, by Baker's theorem on linear forms in logarithms [1], either 0 or transcendental (see [2, 8]; for related transcendence results see [13]); thus Conjecture 1.1 is equivalent to the statement that *every convergent Erdős sum $\sum f(n)/n$ is transcendental.* Okada [10] gave a criterion for $L(1, f) = 0$ as a system of $\varphi(q) + \omega(q)$ rational linear equations and proved the conjecture when $2\varphi(q) + 1 > q$. Murty and Saradha [9] proved it for $q \equiv 3 \pmod 4$. Chatterjee and Murty [4] proved it for a set of $q \equiv 1 \pmod 4$ of lower density at least 0.82, and Pathak [12] showed that it holds "with probability one" over Erdős functions. Murty and Pathak [7] found an explicit basis of the Okada space and remarked that it "does not seem to help" for $q \equiv 1 \pmod 4$; Bharadwaj [3] recast Okada's criterion through primitive characters and obtained a further family of admissible q. Livingston's own proposed route was shown to fail for composite q by Pathak [11]. As far as we are aware the conjecture has been regarded as open for $q \equiv 1 \pmod 4$ (it is stated as open in [12, 7, 3]).

1.2. **Results.**

Theorem A (Theorem 3.1)**.** *Let $q = 11025 = 3^2 \cdot 5^2 \cdot 7^2$. There is an Erdős function f modulo q, given explicitly in Table 1 in terms of Legendre symbols, with $\sum_{n=1}^{q} f(n) = 0$ and*

$$\sum_{n \geq 1} \frac{f(n)}{n} = 0.$$

In particular Conjecture 1.1 is false.

The proof of Theorem A is elementary: by a decomposition of $L(1,f)$ into Dirichlet L-values and logarithms of primes (Proposition 2.1), the vanishing of $L(1,f)$ follows from seven identities between rational numbers attached to the seven non-trivial real characters of conductor dividing 105, three identities attached to $\log 3, \log 5, \log 7$, and the convergence identity; all eleven are listed in Tables 2 and 3. High-precision numerics give $|L(1,f)| < 10^{-130}$ as an independent check.

Theorem B (Theorems 3.3 and 3.6)**.** *There is also an Erdős function modulo* $165^2 = 27225$ *with vanishing sum. If* q *and* N *carry Erdős functions with vanishing sums, so does* qN*. Hence every* $q = 11025^a\, 27225^b$ *with* $a + b \geq 1$ *carries one; so do* 315^2*,* 495^2 *and* 735^2 *(Theorem 3.4).*

The counterexamples were not found by blind search: the space of Erdős functions modulo 11025 has 2^{11024} elements. They come from a 2-adic principle which we now describe. Because $\pm 1 \equiv 1 \pmod 2$, an Erdős function with $L(1,f) = 0$ is a vector of the Okada lattice all of whose coordinates are odd; we call such vectors *odd Okada vectors* (Definition 4.1). Their existence is a question about the 2-adic integral structure of the transcendental relations among the numbers $\psi(a/q)$, and it can be settled by finite linear algebra.

Theorem C (2-shadow descent, Theorem 4.5)**.** *Let* $H \leq (\mathbb{Z}/q\mathbb{Z})^\times$ *be the subgroup of elements of odd order. If* q *has an odd Okada vector, it has one that is invariant under multiplication by* H*. For such vectors Okada's criterion only involves characters of* 2*-power order, and it becomes a linear system on the maximal* 2*-power quotients* $\prod_{p|d} C_{2^{v_2(p-1)}}$ *of the groups* $(\mathbb{Z}/d\mathbb{Z})^\times$*,* $d \mid q$*, with* $\prod_{p^e \| q}(1 + e\, 2^{v_2(p-1)}) - 1$ *unknowns.*

Theorem D (Theorems 5.2, 5.6 and Corollary 5.7)**.**

(1) *Every odd square has an odd Okada vector.*

(2) *No odd non-square* $q \leq 13105$ *has an odd Okada vector. Consequently Conjecture 1.1 holds for every odd non-square* $q \leq 13105$*, and for these* q *every convergent Erdős sum is transcendental. Among them are* 74 *moduli* $q \equiv 1 \pmod 4$ *(the smallest are* 105*,* 165*,* 525*,* 585*,* 825 *and* 945*) not covered by the criteria of* [2, 10, 9, 4, 3].

Part (2) is a computer-assisted proof: for each of the 4938 odd composite non-squares $q \leq 13105$ we exhibit a rational combination of the shadow equations whose coefficients are 2-integral and whose value on the all-ones vector is odd (Section 5 and Appendix B). Together with Theorem A and Okada's theorem this gives a complete picture in a long initial range.

Theorem E (Theorem 5.9)**.** *For* $2 \leq q \leq 13105$*, an Erdős function modulo* q *with convergent and vanishing sum exists if and only if* $q = 11025$*. In particular* 11025 *is the least modulus for which Conjecture 1.1 fails.*

We expect part (2) of Theorem D to hold for all odd non-squares (Conjecture 5.3); this would prove Conjecture 1.1 for all non-square q and confine all counterexamples to square moduli. The shadow system also gives unconditional results for infinite families: a residuacity criterion (Theorem 5.4) proves Conjecture 1.1 whenever some prime $p \parallel q$ has all other prime factors of q as $2^{v_2(p-1)}$-th power residues, which yields infinitely many new $q \equiv 1 \pmod 4$ (for example $q = 3 \cdot 7 \cdot 13 \cdot 19 \cdot 31$); and the same principle gives a four-line proof of the theorem of Murty and Saradha for $q \equiv 3 \pmod 4$ (Proposition 4.4).

1.3. **The idea in one paragraph.** Okada's criterion expresses $L(1,f) = 0$ as a rational linear system; its solution space $\mathcal{O}_q$ is stable under multiplication of the argument by units. An Erdős function with $L(1,f) = 0$ lies in $\mathcal{O}_q$ and is $\equiv 1$ modulo 2 at every non-zero residue. Averaging such a vector over the odd-order part H of the unit group keeps it 2-integral and odd, because $|H|$ is odd, so nothing 2-adic is lost by passing to H-invariant vectors (Theorem C). The remaining system lives on small 2-groups and can be decided exactly. When it has no odd solution (which,

as far as we can test, happens precisely for non-squares) the conjecture follows for q. When it does have odd solutions (squares), one may look for solutions with values ± 1; lifting such a shadow solution back along the fibres of the projection gives an honest Erdős function with $L(1, f) = 0$. This is how the counterexamples of Theorems A and B were found (Section 6).

1.4. **Organization.** Section 2 recalls Okada's criterion in a layered form and proves the decomposition used for Theorem A. Section 3 contains the counterexamples and the semigroup property. Section 4 introduces odd Okada vectors, re-proves the Murty–Saradha theorem, and proves the shadow descent. Section 5 treats squares and non-squares and proves Theorem E. Section 6 explains the search, and Section 7 lists open problems. The appendices contain the second counterexample, the certificate format, and a description of the accompanying code and data.

2. Okada's criterion in layered form

Throughout, $q \geq 3$ is odd, $q = \prod_{p|q} p^{e_p}$, $U_d = (\mathbb{Z}/d\mathbb{Z})^\times$, and φ is Euler's function. For a divisor $d > 1$ of q the *layer* of level d is the set of residues $n \bmod q$ with $\gcd(n, q) = q/d$, that is $n = (q/d)x$ with $x \in U_d$. The non-zero residues modulo q are the disjoint union of the layers, and $\sum_{d|q,\, d>1} \varphi(d) = q - 1$. For a q-periodic f with $f(0) = 0$ we write

$$f_d(x) = f\big((q/d)x\big) \quad (x \in U_d), \qquad S_d = \sum_{x \in U_d} f_d(x), \qquad \langle f_d, \chi \rangle = \sum_{x \in U_d} f_d(x)\overline{\chi(x)}$$

for Dirichlet characters χ modulo d. A character χ modulo d is induced by a unique primitive character χ^* whose conductor f_χ divides d. For a prime $p \mid q$ and a layer d put

$$\varepsilon_p(d) = v_p(q/d) - \frac{[\,p \mid d\,]}{p-1}. \tag{2.1}$$

Proposition 2.1 (Decomposition)**.** *Let f be rational-valued, q-periodic, with $f(0) = 0$ and $\sum_{n=1}^{q} f(n) = \sum_d S_d = 0$. Then*

$$L(1, f) = \sum_{\chi^*} A_{\chi^*}(f)\, L(1, \chi^*) \;-\; \frac{1}{q} \sum_{p|q} \Big(\sum_{d|q,\, d>1} S_d\, \varepsilon_p(d) \Big) \log p, \tag{2.2}$$

where χ^* runs over the non-principal primitive characters whose conductor divides q, and

$$A_{\chi^*}(f) = \frac{1}{q} \sum_{\substack{d|q \\ f_\chi | d}} \frac{d}{\varphi(d)} \prod_{\substack{p|d \\ p \nmid f_\chi}} \Big(1 - \frac{\chi^*(p)}{p}\Big) \langle f_d, \chi^* \rangle. \tag{2.3}$$

Proof. For $s > 1$ group the terms of $L(s, f) = \sum f(n) n^{-s}$ according to $\gcd(n, q) = q/d$; writing $n = (q/d)x$ with $\gcd(x, d) = 1$ and expanding f_d in characters modulo d,

$$L(s, f) = \sum_d \Big(\frac{d}{q}\Big)^s \sum_{\gcd(x,d)=1} \frac{f_d(x)}{x^s} = \sum_d \Big(\frac{d}{q}\Big)^s \frac{1}{\varphi(d)} \sum_{\chi \bmod d} \langle f_d, \chi \rangle L(s, \chi).$$

For non-principal χ, $L(s, \chi) = L(s, \chi^*) \prod_{p|d,\, p \nmid f_\chi} (1 - \chi^*(p) p^{-s})$ is continuous at $s = 1$, and collecting the terms with a given χ^* gives the first sum in (2.2). For the principal character, $L(s, \chi_0) = \zeta(s) \prod_{p|d} (1 - p^{-s})$, and as $s \to 1^+$

$$\Big(\frac{d}{q}\Big)^s \frac{S_d}{\varphi(d)} \zeta(s) \prod_{p|d} (1 - p^{-s}) = \frac{S_d}{q} \Big(\frac{1}{s-1} + \gamma - \log \frac{q}{d} + \sum_{p|d} \frac{\log p}{p-1} + O(s-1) \Big).$$

Summing over d, the pole and the γ-terms cancel because $\sum_d S_d = 0$, and what remains is $-\frac{1}{q}\sum_d S_d \sum_p \varepsilon_p(d)\log p$. Since the Dirichlet series converges at $s=1$, $L(1,f) = \lim_{s\to 1^+} L(s,f)$ by Abel's theorem. □

The coefficient of $\log p$ can also be written $-\frac{1}{q}\sum_{n=1}^{q-1} f(n)\varepsilon_p(n)$ with $\varepsilon_p(n) = v_p(\gcd(n,q)) - [v_p(\gcd(n,q)) < e_p]/(p-1)$. Proposition 2.1 immediately gives the sufficiency half of the following criterion; necessity rests on Baker's theorem.

Theorem 2.2 (Okada [10]; layered form, cf. Bharadwaj [3, Thm. 4])**.** *Let f be as in Proposition 2.1. Then $L(1,f) = 0$ if and only if*

(A) $A_{\chi^*}(f) = 0$ *for every non-principal primitive χ^* of conductor dividing q, and*
(B) $\sum_d S_d\,\varepsilon_p(d) = 0$ *for every prime $p \mid q$.*

Sketch of necessity. For odd χ^* one has $L(1,\chi^*) \in \pi i\,\overline{\mathbb{Q}}$, and for even $\chi^* \neq 1$ one has $L(1,\chi^*) = -\frac{\tau(\chi^*)}{f_\chi}\sum_a \overline{\chi^*}(a)\log|1-\zeta_{f_\chi}^a|$. The number $\pi i = \log(-1)$ is $\mathbb{Q}$-linearly independent of logarithms of positive real algebraic numbers, so by Baker's theorem the π-part and the real-logarithmic part of (2.2) vanish separately. By Gauss's formula for $\psi(a/q)$ the π-part is $\frac{\pi}{2q}\sum_a f(a)\cot(\pi a/q) = -\frac{\pi i}{2q}X$ with $X = \sum_a f(a)\frac{1+\zeta_q^a}{1-\zeta_q^a} \in \mathbb{Q}(\zeta_q)$ (here f is rational), and $X = 0$ if and only if all its Galois conjugates $\sum_a f(a)\frac{1+\zeta_q^{ca}}{1-\zeta_q^{ca}}$, $c \in U_q$, vanish; since replacing f by $f(c^{-1}\cdot)$ multiplies A_{χ^*} by $\overline{\chi^*}(c)$, orthogonality and $L(1,\chi^*) \neq 0$ give $A_{\chi^*} = 0$ for odd χ^*. For the logarithmic part one decomposes the $\overline{\mathbb{Q}}$-span of the classes of $\log|1-\zeta^a|$ and $\log p$ into eigenspaces of $\mathrm{Gal}(\mathbb{Q}(\zeta_q)/\mathbb{Q})$: the even $L(1,\chi^*)$ are non-zero vectors in pairwise distinct eigenspaces, and the classes of $\log p$, $p \mid q$, are linearly independent in the invariant eigenspace; Baker's theorem turns this into linear independence of the corresponding numbers over $\overline{\mathbb{Q}}$. This is the argument of [2, 10]; in the present layered form it is [3, Thm. 4]. □

Theorem 2.2, together with $\sum_d S_d = 0$, is a system of $\varphi(q)+\omega(q)$ independent rational linear equations in the $q-1$ values $f(1),\dots,f(q-1)$ (the characters contribute $\varphi(q)-1$ real equations, (B) contributes $\omega(q)$). It is equivalent to Okada's original system [10]: $\sum_{m\in M(q)} f(am)/m = 0$ for all $a \in U_q$, where $M(q)$ is the multiplicative monoid generated by the primes dividing q, together with one equation for each $p \mid q$. We denote by

$$\mathcal{O}_q = \Big\{ f\colon (\mathbb{Z}/q\mathbb{Z}) \smallsetminus \{0\} \to \mathbb{Q} \;:\; \textstyle\sum_n f(n) = 0 \text{ and (A), (B) hold} \Big\}$$

the solution space (with $f(0) = 0$); $\dim \mathcal{O}_q = q - 1 - \varphi(q) - \omega(q)$. It is stable under $f \mapsto f(u\,\cdot)$ for $u \in U_q$, because this multiplies each $A_{\chi^*}(f)$ by a root of unity and leaves all S_d unchanged.

3. Counterexamples

3.1. **The modulus 105^2.** Let $q = 11025 = 3^2\cdot 5^2\cdot 7^2$. For $n \not\equiv 0 \pmod q$ write $n = (q/d)x$ with $d \mid q$, $d > 1$, $x \in U_d$, and for $p \mid d$ put $\lambda_p = \left(\frac{x}{p}\right)$ (Legendre symbol). Define $f(n) = F_d(x)$ with F_d as in Table 1 and $f(n) = 0$ for $q \mid n$. Each F_d is a ± 1-valued function of the Legendre symbols λ_p, $p \mid d$, so f is an Erdős function modulo q. For $m \mid 105$ let χ_m be the real primitive character of conductor m, $\chi_m(x) = \prod_{p\mid m}\left(\frac{x}{p}\right)$, and write $\lambda_m = \chi_m(x)$. The last column of Table 1 records the expansion $F_d = \sum_{m\mid \mathrm{rad}(d)} c_{d,m}\lambda_m$.

Table 1: The counterexample modulo $q = 11025$: for n with $\gcd(n,q) = q/d$ write $n = (q/d)x$; then $f(n) = F_d(x)$, where $\lambda_p = \left(\frac{x}{p}\right)$. The last column gives the expansion $F_d = \sum_m c_{d,m}\lambda_m$ ($\lambda_m = \prod_{p\mid m}\lambda_p$).

d	q/d	$\varphi(d)$	$F_d(x)$	nonzero $c_{d,m}$
3	3675	2	-1	$c_1 = -1$
5	2205	4	$-\lambda_5$	$c_5 = -1$
7	1575	6	$-\lambda_7$	$c_7 = -1$
9	1225	6	-1	$c_1 = -1$
15	735	8	$-\lambda_3\lambda_5$	$c_{15} = -1$
21	525	12	1 if $\lambda_3 = 1$; $-\lambda_7$ if $\lambda_3 = -1$	$c_1 = \frac{1}{2}$, $c_7 = -\frac{1}{2}$, $c_3 = \frac{1}{2}$, $c_{21} = \frac{1}{2}$
25	441	20	λ_5	$c_5 = 1$
35	315	24	$\lambda_5\lambda_7$	$c_{35} = 1$
45	245	24	-1 if $\lambda_3 = 1$; λ_5 if $\lambda_3 = -1$	$c_1 = -\frac{1}{2}$, $c_5 = \frac{1}{2}$, $c_3 = -\frac{1}{2}$, $c_{15} = -\frac{1}{2}$
49	225	42	$-\lambda_7$	$c_7 = -1$
63	175	36	1 if $\lambda_3 = 1$; $-\lambda_7$ if $\lambda_3 = -1$	$c_1 = \frac{1}{2}$, $c_7 = -\frac{1}{2}$, $c_3 = \frac{1}{2}$, $c_{21} = \frac{1}{2}$
75	147	40	1 if $\lambda_3 = 1$; $-\lambda_5$ if $\lambda_3 = -1$	$c_1 = \frac{1}{2}$, $c_5 = -\frac{1}{2}$, $c_3 = \frac{1}{2}$, $c_{15} = \frac{1}{2}$
105	105	48	$-\lambda_7$ if $\lambda_3 = 1$; -1 if $\lambda_3 = -1$	$c_1 = -\frac{1}{2}$, $c_7 = -\frac{1}{2}$, $c_3 = \frac{1}{2}$, $c_{21} = -\frac{1}{2}$
147	75	84	λ_7	$c_7 = 1$
175	63	120	$-\lambda_5\lambda_7$	$c_{35} = -1$
225	49	120	λ_3	$c_3 = 1$
245	45	168	$\lambda_5\lambda_7$	$c_{35} = 1$
315	35	144	$\lambda_5\lambda_7$	$c_{35} = 1$
441	25	252	$-\lambda_3$	$c_3 = -1$
525	21	240	$\lambda_5\lambda_7$	$c_{35} = 1$
735	15	336	$\lambda_5\lambda_7$	$c_{35} = 1$
1225	9	840	$\lambda_5\lambda_7$	$c_{35} = 1$
1575	7	720	$\lambda_5\lambda_7$	$c_{35} = 1$
2205	5	1008	$\lambda_5\lambda_7$	$c_{35} = 1$
3675	3	1680	$\lambda_5\lambda_7$	$c_{35} = 1$
11025	1	5040	$-\lambda_5\lambda_7$	$c_{35} = -1$

Theorem 3.1. *The function f of Table 1 satisfies $\sum_{n=1}^{q} f(n) = 0$ and*

$$\sum_{n=1}^{\infty} \frac{f(n)}{n} = 0.$$

Proof. We apply Proposition 2.1. Since each F_d depends only on Legendre symbols, $F_d(xy^2) = F_d(x)$ for all $x, y \in U_d$, hence $\langle f_d, \chi\rangle = \overline{\chi(y)}^{\,2}\langle f_d, \chi\rangle$ for all y; so $\langle f_d, \chi\rangle = 0$ unless $\chi^2 = 1$. A real primitive character of odd conductor dividing q has squarefree conductor, so it is one of χ_m, $m \mid 105$, $m > 1$. By orthogonality of the distinct characters λ_m on U_d, $\langle f_d, \chi_m\rangle = \varphi(d)\, c_{d,m}$ and $S_d = \varphi(d)\, c_{d,1}$. Hence (2.3) gives

$$q\, A_{\chi_m}(f) = \sum_{m\mid d\mid q} w_{d,m}\, c_{d,m}, \qquad w_{d,m} = d \prod_{\substack{p\mid d \\ p\nmid m}} \Big(1 - \frac{\chi_m(p)}{p}\Big),$$

and all other coefficients $A_{\chi^*}(f)$ vanish. The values $\chi_m(p)$ follow from $\left(\frac{3}{5}\right) = \left(\frac{3}{7}\right) = \left(\frac{5}{3}\right) = \left(\frac{5}{7}\right) = \left(\frac{7}{5}\right) = -1$ and $\left(\frac{7}{3}\right) = 1$. Table 2 lists every non-zero term $w_{d,m}c_{d,m}$; each row sums to 0 (for $m = 105$ there is no term). Table 3 lists the non-zero layer sums S_d; they sum to 0, which is the convergence condition, and $\sum_d S_d \varepsilon_p(d) = 0$ for $p = 3, 5, 7$. By (2.2), $L(1, f) = 0$. □

TABLE 2. The seven identities $\sum_{m|d} w_{d,m}c_{d,m} = 0$, with $w_{d,m} = d\prod_{p|d,\, p\nmid m}\big(1 - \chi_m(p)/p\big)$; entries are $w_{d,m}c_{d,m}$ for the layers with $c_{d,m} \neq 0$.

m	$\chi_m(p)$, $p \nmid m$	terms $w_{d,m}c_{d,m}$ (layer d in brackets)
3	$\chi_3(5) = -1$, $\chi_3(7) = +1$	$9\,[21]$ $-27\,[45]$ $27\,[63]$ $45\,[75]$ $54\,[105]$ $270\,[225]$ $-378\,[441]$
5	$\chi_5(3) = -1$, $\chi_5(7) = -1$	$-5\,[5]$ $25\,[25]$ $30\,[45]$ $-50\,[75]$
7	$\chi_7(3) = -1$, $\chi_7(5) = -1$	$-7\,[7]$ $-14\,[21]$ $-49\,[49]$ $-42\,[63]$ $-84\,[105]$ $196\,[147]$
15	$\chi_{15}(7) = -1$	$-15\,[15]$ $-\frac{45}{2}\,[45]$ $\frac{75}{2}\,[75]$
21	$\chi_{21}(5) = +1$	$\frac{21}{2}\,[21]$ $\frac{63}{2}\,[63]$ $-42\,[105]$
35	$\chi_{35}(3) = +1$	$35\,[35]$ $-175\,[175]$ $245\,[245]$ $210\,[315]$ $350\,[525]$ $490\,[735]$ $1225\,[1225]$ $1050\,[1575]$ $1470\,[2205]$ $2450\,[3675]$ $-7350\,[11025]$
105	—	

TABLE 3. Layer sums $S_d = \varphi(d)c_{d,1}$ (layers with $S_d \neq 0$) and the weights $\varepsilon_p(d) = v_p(q/d) - [p \mid d]/(p-1)$.

d	S_d	$\varepsilon_3(d)$	$\varepsilon_5(d)$	$\varepsilon_7(d)$
3	-2	$\frac{1}{2}$	2	2
9	-6	$-\frac{1}{2}$	2	2
21	6	$\frac{1}{2}$	2	$\frac{5}{6}$
45	-12	$-\frac{1}{2}$	$\frac{3}{4}$	2
63	18	$-\frac{1}{2}$	2	$\frac{5}{6}$
75	20	$\frac{1}{2}$	$-\frac{1}{4}$	2
105	-24	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{5}{6}$
$\sum$	0	$\sum S_d\varepsilon_3 = 0$	$\sum S_d\varepsilon_5 = 0$	$\sum S_d\varepsilon_7 = 0$

Remark 3.2. The theorem was checked in three further ways: (i) all 5040 of Okada's unit equations $\sum_{m\in M(q)} f(am)/m = 0$ and his three prime equations hold exactly in rational arithmetic, with the infinite sums evaluated as finite combinations of geometric series; (ii) evaluating $L(1, f) = -\frac{1}{q}\sum_{a=1}^{q-1} f(a)\psi(a/q)$ in PARI/GP [16] with 134 significant digits gives $L(1, f) \approx -1.2 \cdot 10^{-135}$, i.e. zero to working precision, while exchanging the two values $f(1) = -1$ and $f(2) = 1$ changes it to $1.00000003\ldots$; (iii) the partial sum $\sum_{n\le 5q} f(n)/n \approx 2.3 \cdot 10^{-5}$ is of the size expected for a convergent series with value 0.

3.2. The modulus 165^2.

Theorem 3.3. *Let $q = 27225 = 3^2 \cdot 5^2 \cdot 11^2$ and let f be the Erdős function modulo q given in Table 4 (Appendix A). Then $\sum_{n=1}^{q} f(n) = 0$ and $L(1, f) = 0$.*

Proof (computer-verified). The function f is invariant under the subgroup $H = \{u^{16} : u \in U_q\}$ of odd-order units ($|H| = 825$), so $\langle f_d, \chi\rangle = 0$ unless χ has 2-power order. The primitive characters of 2-power order and conductor dividing q are $\chi = \left(\frac{\cdot}{3}\right)^a \omega_5^b \left(\frac{\cdot}{11}\right)^c$ with $a, c \in \{0, 1\}$, $b \in \{0, 1, 2, 3\}$, where ω_5 is the quartic character modulo 5 with $\omega_5(2) = i$. For each of the 15 non-trivial ones the

coefficient (2.3) was computed exactly in $\mathbb{Q}(i)$ and found to be 0; the three sums $\sum_d S_d\varepsilon_p(d)$, $p \in \{3,5,11\}$, and $\sum_d S_d$ are 0. As independent checks, all 13200 unit equations and the three prime equations of Okada's original system hold exactly, and PARI/GP gives $L(1,f) \approx -1.1 \cdot 10^{-135}$ with 134 significant digits. □

Unlike the first example, no counterexample modulo 27225 that is invariant under H depends only on Legendre symbols (the corresponding finite search space is empty, Section 6); the quartic character modulo 5 is needed.

3.3. **Further examples: 315^2, 495^2 and 735^2.** Restricting the search of Section 6 to H-invariant functions that depend only on Legendre symbols also produced counterexamples modulo

$$315^2 = 3^4 \cdot 5^2 \cdot 7^2 = 99225, \qquad 495^2 = 3^4 \cdot 5^2 \cdot 11^2 = 245025, \qquad 735^2 = 3^2 \cdot 5^2 \cdot 7^4 = 540225.$$

Theorem 3.4. *For $q \in \{315^2, 495^2, 735^2\}$ there is an Erdős function modulo q with vanishing sum which depends, on every layer, only on the Legendre symbols $\left(\frac{x}{p}\right)$, $p \mid q$.*

Proof. For a function of this type the proof of Theorem 3.1 applies verbatim: only the real characters χ_m, $m \mid \mathrm{rad}(q)$, can occur, and $L(1,f) = 0$ follows from the seven identities $\sum_{m|d} w_{d,m}c_{d,m} = 0$, the three logarithmic identities and the convergence identity, now summed over the 44 layers of q. For the three functions in the supplementary material all eleven identities hold exactly (script `verify_legendre_type.py`). □

For 99225 we also checked all 45360 unit equations and the three prime equations of Okada's original system exactly, and PARI/GP gives $|L(1,f)| \approx 1.2 \cdot 10^{-135}$ with 134 significant digits; for 245025 and 540225, PARI/GP gives $|L(1,f)| \approx 7.8 \cdot 10^{-78}$ and $1.5 \cdot 10^{-77}$ with 77 significant digits. These examples do not follow from Theorem 3.6, since 9 and 49 carry no counterexamples (Okada's theorem); note that $495^2 = 9 \cdot 165^2$ has a Legendre-type counterexample although 165^2 has no H-invariant one of Legendre type.

3.4. **Products of counterexample moduli.**

Lemma 3.5 (Composition)**.** *Let f be q-periodic and g be N-periodic, rational-valued, with $f(0) = g(0) = 0$ and $\sum_{n=1}^{q} f(n) = \sum_{n=1}^{N} g(n) = 0$. Define the qN-periodic function*

$$F(n) = \begin{cases} f(n), & q \nmid n, \\ g(n/q), & q \mid n. \end{cases}$$

Then $F(0) = 0$, $\sum_{n=1}^{qN} F(n) = 0$, and $L(1,F) = L(1,f) + \frac{1}{q}L(1,g)$. If f and g are Erdős functions modulo q and N, then F is an Erdős function modulo qN.

Proof. Periodicity and $F(0) = g(0) = 0$ are clear. Over a period, $\sum_{q\nmid n} F(n) = N\sum_{n=1}^{q} f(n) = 0$ and $\sum_{q|n} F(n) = \sum_{m=1}^{N} g(m) = 0$. The series $\sum_{q\nmid n} f(n)/n = L(1,f)$ (recall $f(n) = 0$ for $q \mid n$) and $\sum_m g(m)/(qm) = L(1,g)/q$ both converge, and their sum is $L(1,F)$. If $f(n) = \pm 1$ for $q \nmid n$ and $g(m) = \pm 1$ for $N \nmid m$, then $F(n) = \pm 1$ unless $q \mid n$ and $N \mid n/q$, i.e. unless $qN \mid n$. □

Theorem 3.6. *Let $\mathcal{C}$ be the set of moduli q for which an Erdős function with $\sum_{n=1}^{q} f(n) = 0$ and $L(1,f) = 0$ exists. Then $\mathcal{C}$ is closed under multiplication. In particular*

$$105^{2a}\,165^{2b}\,315^{2c}\,495^{2e}\,735^{2g} \in \mathcal{C} \qquad (a,b,c,e,g \geq 0 \text{ not all } 0).$$

Proof. Apply Lemma 3.5 with $L(1,f) = L(1,g) = 0$, and use Theorems 3.1, 3.3 and 3.4. □

4. The 2-adic principle

Let $\mathbb{Z}_{(2)}$ be the ring of rationals with odd denominator. Every element of $\mathbb{Z}_{(2)}$ has a residue modulo 2, and we call it odd or even accordingly.

Definition 4.1. An *odd Okada vector* for q is an $f \in \mathcal{O}_q$ with $f(n) \in \mathbb{Z}_{(2)}$ and $f(n)$ odd for every $n \not\equiv 0 \pmod q$. We say that q is 2-*adically obstructed* if no odd Okada vector exists.

Lemma 4.2. *If f is an Erdős function modulo q with $\sum_{n=1}^{q} f(n) = 0$ and $L(1, f) = 0$, then f is an odd Okada vector. In particular, Conjecture 1.1 holds for every 2-adically obstructed q, and for such q every convergent Erdős sum is transcendental.*

Proof. By Theorem 2.2, $f \in \mathcal{O}_q$, and ± 1 is odd. The last statement follows from the fact that $L(1, f)$ is 0 or transcendental [2, 8]. □

The search for odd Okada vectors is a linear problem over the discrete valuation ring $\mathbb{Z}_{(2)}$, and it has a dual description.

Lemma 4.3 (Certificates)**.** *Let $R_1, \dots, R_k \in \mathbb{Q}^N$ span the space of linear forms vanishing on a subspace $V \subseteq \mathbb{Q}^N$, and let $\mathbf{1} = (1, \dots, 1)$. There is $v \in V \cap \mathbb{Z}_{(2)}^N$ with all coordinates odd if and only if there is no $y \in \mathbb{Q}^k$ such that $\Lambda = \sum_i y_i R_i$ has all coordinates in $\mathbb{Z}_{(2)}$ and $\Lambda(\mathbf{1})$ is odd.*

Proof. Let $K = V \cap \mathbb{Z}_{(2)}^N$ and $R = V^\perp \cap \mathbb{Z}_{(2)}^N$; both are saturated $\mathbb{Z}_{(2)}$-lattices, of ranks $\dim V$ and $N - \dim V$. Their reductions $\overline{K}, \overline{R} \subseteq \mathbb{F}_2^N$ have dimensions $\dim V$ and $N - \dim V$ (a saturated lattice has a basis that reduces to an $\mathbb{F}_2$-basis) and are orthogonal, hence $\overline{R} = \overline{K}^\perp$. An odd vector in K exists iff $\mathbf{1} \in \overline{K} = \overline{R}^\perp$, i.e. iff $\Lambda(\mathbf{1})$ is even for every $\Lambda \in R$. □

4.1. **A four-line proof for $q \equiv 3 \pmod 4$.** The simplest instance of the principle recovers the theorem of Murty and Saradha [9]. Put $\zeta = e^{2\pi i/q}$ and $\rho_a = \zeta^a/(1 - \zeta^a)$. Since q is odd, $1 - \zeta^a$ is a unit at every prime above 2 for $a \not\equiv 0$, so ρ_a is 2-integral.

Proposition 4.4. *If $q \equiv 3 \pmod 4$, then q is 2-adically obstructed. In particular Conjecture 1.1 holds for $q \equiv 3 \pmod 4$ [9].*

Proof. Let $f \in \mathcal{O}_q$. The vanishing of the π-part of (2.2) (see the proof of Theorem 2.2) reads

$$\sum_{a=1}^{q-1} f(a) \cot(\pi a/q) = 0;$$

since

$$i \cot(\pi a/q) = \frac{1 + \zeta^a}{1 - \zeta^a} = 1 + 2\rho_a$$

and $\sum_a f(a) = 0$, this is $\sum_a f(a)\rho_a = 0$. Pairing a with $-a$ and using $\rho_a + \rho_{-a} = -1$ gives $\sum_{a=1}^{q-1} \rho_a = -(q-1)/2$. If f were odd and 2-integral, then $f(a) = 1 + 2g(a)$ with $g(a) \in \mathbb{Z}_{(2)}$ and

$$0 = \sum_a f(a)\rho_a = -\frac{q-1}{2} + 2\sum_a g(a)\rho_a,$$

which is impossible when $(q-1)/2$ is odd. □

For $q \equiv 1 \pmod 4$ the same identity only says $\sum_a g(a)\rho_a = (q-1)/4$, which is not contradictory by itself; one needs the interaction of the odd and even parts of Okada's system, which is what the following descent makes computable.

4.2. **Descent to the 2-shadow.** For $d \mid q$ let $H_d \le U_d$ be the subgroup of elements of odd order and $Q_d = U_d/H_d$, the maximal 2-power quotient. For an odd prime p let $t_p = v_2(p-1)$. Since U_{p^e} is cyclic of order $p^{e-1}(p-1)$, the 2-part of U_{p^e} maps isomorphically onto that of U_p, and

$$Q_d \;\cong\; \prod_{p \mid d} C_{2^{t_p}}, \qquad x \mapsto \big(\mathrm{ind}_{g_p}(x \bmod p) \bmod 2^{t_p}\big)_{p \mid d},$$

where g_p is a primitive root modulo p. In particular Q_d depends only on $\mathrm{rad}(d)$, and $|H_d| = \varphi(d)/|Q_d|$ is odd. The characters of Q_d are exactly the characters of 2-power order modulo d; their conductors are squarefree, and the primitive ones of conductor $m \mid \mathrm{rad}(q)$ are the characters of Q_m that are non-trivial on every factor $C_{2^{t_p}}$, $p \mid m$.

Theorem 4.5 (2-shadow descent)**.** *Let q be odd and $H = H_q$.*

1. *If q has an odd Okada vector, then it has one that is invariant under $n \mapsto hn$ for all $h \in H$.*
2. *An H-invariant f is determined by the functions $\Phi_d \colon Q_d \to \mathbb{Q}$, $\Phi_d(y) = \sum_{x \mapsto y} f_d(x)$ (fibre sums over $U_d \to Q_d$), and f is odd and 2-integral iff every Φ_d is.*
3. *For H-invariant f, $f \in \mathcal{O}_q$ if and only if the following hold.*

   (A′) *For every $m \mid \mathrm{rad}(q)$, $m > 1$, and every primitive $\chi \in \widehat{Q_m}$:*

$$\sum_{m \mid d \mid q} \frac{d}{\varphi(d)} \prod_{p \mid d,\, p \nmid m} \Big(1 - \frac{\chi(p)}{p}\Big) \sum_{y \in Q_d} \Phi_d(y)\overline{\chi(y)} = 0.$$

   (B′) *For every $p \mid q$: $\sum_d \varepsilon_p(d) \sum_y \Phi_d(y) = 0$.*

   (C′) $\sum_d \sum_y \Phi_d(y) = 0$.

Consequently, q is 2-adically obstructed if and only if the linear system (A′)–(C′) *has no solution with all $\Phi_d(y)$ odd and 2-integral. The number of unknowns is $\sum_{d \mid q,\, d>1} |Q_d| = \prod_{p^e \| q} \big(1 + e\, 2^{t_p}\big) - 1$.*

Proof. (1) Let f be an odd Okada vector and $\bar f(n) = |H|^{-1} \sum_{h \in H} f(hn)$. Then $\bar f \in \mathcal{O}_q$ because $\mathcal{O}_q$ is stable under multiplication by units (Section 2); $\bar f$ is 2-integral because $|H|$ is odd; and $\bar f(n)$ is odd, since $\sum_h f(hn)$ is a sum of an odd number of odd elements of $\mathbb{Z}_{(2)}$. Clearly $\bar f$ is H-invariant.

(2) The image of H in U_d is H_d (a surjection of finite abelian groups maps the odd-order part onto the odd-order part), so an H-invariant f has f_d constant on the fibres of $U_d \to Q_d$, which have odd cardinality $|H_d|$; hence $f_d = \Phi_d/|H_d|$ on each fibre and the parity statement follows.

(3) For H-invariant f and a character χ modulo d, $\langle f_d, \chi\rangle = 0$ unless χ is trivial on H_d, i.e. unless χ has 2-power order, in which case $\langle f_d, \chi\rangle = \sum_y \Phi_d(y)\overline{\chi(y)}$. So in Theorem 2.2 the conditions (A) for characters not of 2-power order hold automatically, and the remaining ones are (A′); (B′) and (C′) are (B) and the convergence condition, as $S_d = \sum_y \Phi_d(y)$. The count of unknowns follows from $|Q_d| = \prod_{p \mid d} 2^{t_p}$ and the fact that there are e_p choices of $v_p(d) \ge 1$ for each $p \mid d$. □

Remark 4.6. For computations we use an equivalent integer form of (A′). For $m \mid \mathrm{rad}(q)$, $m > 1$, and $y \in Q_m$ put

$$H_m(\Phi)(y) = \sum_{m \mid d \mid q} \Big(\prod_{p \mid q,\, p \nmid d} (p-1) \Big) \Big[\prod_{p \mid d,\, p \nmid m} \big(p - T_p\big)\, \pi_{d \to m} \Phi_d \Big](y),$$

where $\pi_{d \to m}$ sums over the fibres of $Q_d \to Q_m$ and T_p is translation by the image of p in Q_m. Multiplying (A′) by the non-zero constant $\frac{\varphi(m)}{m} \prod_{p \mid q,\, p \nmid m} (p-1)$ shows that (A′) for all primitive

$\chi \in \widehat{Q_m}$ is equivalent to: $H_m(\Phi)$ has no primitive component, i.e.

$$\sum_{\epsilon\in\{0,1\}^{\omega(m)}} (-1)^{|\epsilon|}\, H_m(\Phi)\big(y_\epsilon(k)\big) = 0 \qquad \text{for all } k=(k_p)\in \prod_{p\mid m}\big(C_{2^{t_p}}\smallsetminus\{0\}\big),$$

where $y_\epsilon(k)_p = k_p$ if $\epsilon_p = 0$ and $y_\epsilon(k)_p = 0$ if $\epsilon_p = 1$. Together with (B′) multiplied by $p-1$ and (C′), these are integer linear forms in the unknowns $\Phi_d(y)$.

Remark 4.7. The decisive feature of Theorem 4.5 is that *nothing* 2*-adic is lost* in passing to the shadow: the odd-order part of the Galois group $\mathrm{Gal}(\mathbb{Q}(\zeta_q)/\mathbb{Q}) \cong U_q$ can be averaged away exactly because its order is odd, and what remains is governed by the 2-power cyclotomic subfields. The size of the problem drops from $q-1$ unknowns to $\prod(1+e_p 2^{t_p})-1$; for $q = 11025$ from 11024 to 224. Relaxations that keep only the quadratic characters, or only characters supported on few primes, or only the odd (resp. even) characters, do *not* suffice in general (Section 6); the full 2-power quotient is needed.

5. Squares and non-squares

5.1. **Squares are not obstructed.**

Proposition 5.1. *Let p be an odd prime, $q = p^2$, and $\chi_p = \left(\frac{\cdot}{p}\right)$. Define $f(pa) = \chi_p(a)$ for $p \nmid a$, $f(n) = -\chi_p(n)/p$ for $p \nmid n$, and $f(0) = 0$. Then f is an odd Okada vector for p^2; in fact $L(1, f) = 0$.*

Proof. The values ± 1 and $\mp 1/p$ are odd elements of $\mathbb{Z}_{(2)}$, and $\sum_{n=1}^{p^2} f(n) = 0$ because χ_p sums to 0 over any complete set of units. Moreover

$$L(1,f) = \sum_{p\nmid a} \frac{\chi_p(a)}{pa} - \frac{1}{p}\sum_{p\nmid n}\frac{\chi_p(n)}{n} = \frac{L(1,\chi_p)}{p} - \frac{L(1,\chi_p)}{p} = 0. \qquad \square$$

Theorem 5.2. *Every odd square $q > 1$ has an odd Okada vector.*

Proof. Write $q = p_1^2 p_2^2 \cdots p_r^2$ with primes p_i (repetitions allowed). Lemma 3.5 applies verbatim to odd Okada vectors: if $f \in \mathcal{O}_q$ and $g \in \mathcal{O}_N$ are odd and 2-integral, the function F of the lemma has $F(n) \in \{f(n), g(n/q)\}$ odd and 2-integral for $qN \nmid n$, $\sum F = 0$, and $L(1, F) = L(1, f) + L(1, g)/q = 0$, so $F \in \mathcal{O}_{qN}$ by Theorem 2.2. Starting from Proposition 5.1 and composing $r-1$ times gives an odd Okada vector for q. $\square$

Thus the 2-adic method can never exclude a square modulus, and squares are the only place where it can fail to prove Conjecture 1.1 if the following conjecture holds.

5.2. **Non-squares.**

Conjecture 5.3. *Every odd non-square q is 2-adically obstructed. Equivalently: an odd $q > 1$ has an odd Okada vector if and only if q is a perfect square.*

Conjecture 5.3 implies Conjecture 1.1 for all non-square q, so it would confine all counterexamples to square moduli. It holds for odd prime powers $q = p^e$ with e odd: the layers are $d = p^j$ $(1 \le j \le e)$, all with $Q_d = C_{2^{t_p}}$, the only conductor is $m = p$, and (A′) says that $\Gamma = \sum_j \Phi_{p^j}$ has vanishing non-trivial Fourier coefficients, i.e. is constant; if all $\Phi_{p^j}(y)$ are odd, the constant is a sum of e odd numbers, hence odd, and then $\sum_d S_d = 2^{t_p}\cdot(\text{odd}) \neq 0$ contradicts (C′). For $q \equiv 3 \pmod 4$ it is Proposition 4.4. The shadow system also yields a residuacity criterion that covers infinitely many new moduli $q \equiv 1 \pmod 4$.

Theorem 5.4 (Residuacity criterion)**.** *Let q be odd and let p be a prime with $p \parallel q$. Put $t = v_2(p-1)$. If every prime $r \neq p$ dividing q is a 2^t-th power residue modulo p, then q is 2-adically obstructed; in particular Conjecture 1.1 holds for q.*

Proof. Suppose Φ is an odd 2-integral solution of the shadow system, and write $\Phi_d = |H_d|\mathbf{1} - 2B_d$ with $B_d\colon Q_d \to \mathbb{Z}_{(2)}$ (possible since Φ_d and $|H_d|$ are odd). Let χ be a non-trivial character of $Q_p = C_{2^t}$, viewed as a primitive character of conductor p. By hypothesis $\chi(r) = 1$ for every prime $r \mid q$, $r \neq p$, so in (A′) every weight equals $\frac{d}{\varphi(d)}\prod_{r\mid d,\, r\neq p}(1-\frac{1}{r}) = \frac{p}{p-1}$, and (A′) becomes $\sum_{p\mid d}\sum_y \Phi_d(y)\overline{\chi(y)} = 0$, i.e. $\sum_{p\mid d}\langle B_d, \chi\rangle = 0$. Hence the function $\Gamma = \sum_{p\mid d}\pi_{d\to p}B_d$ on C_{2^t} has vanishing non-trivial Fourier coefficients, so it is a constant c, and $c = \Gamma(0) \in \mathbb{Z}_{(2)}$. Therefore

$$\sum_{p\mid d}\sum_y B_d(y) = 2^t c \equiv 0 \pmod{2^t}.$$

On the other hand, since $p \parallel q$ we have $\varepsilon_p(d) = 1$ for $p \nmid d$ and $\varepsilon_p(d) = -1/(p-1)$ for $p \mid d$, so (B′) and (C′) give $\sum_{p\mid d} S_d = 0$, where $S_d = \sum_y \Phi_d(y) = \varphi(d) - 2\sum_y B_d(y)$. Since $\sum_{p\mid d}\varphi(d) = q - q/p$, this yields $\sum_{p\mid d}\sum_y B_d(y) = \frac{q}{p}\cdot\frac{p-1}{2}$, whose 2-adic valuation is $t-1$, a contradiction. □

Corollary 5.5. *There are infinitely many $q \equiv 1 \pmod 4$ with $2\varphi(q)+1 \le q$ (so that Okada's criterion does not apply) for which Conjecture 1.1 holds; for instance $q = 3\cdot 7\cdot 13\cdot 19\cdot 31 = 160797$.*

Proof. Take $p = 3$ and $q = 3\,r_1\cdots r_k$ with distinct primes $r_i \equiv 1 \pmod 3$; every r_i is then a square modulo 3. Because $\sum_{r\equiv 1\,(3)} 1/r$ diverges, $\varphi(q)/q < \frac{1}{2}$ once k is large, and the condition $q \equiv 1 \pmod 4$ can be met by including or omitting one prime $r \equiv 7 \pmod{12}$. For 160797 one has $\varphi(q)/q \approx 0.484$; this modulus is also not covered by [4, Prop. 3.1, Cor. 3.3] or [3, Cor. 1]. □

Among the 74 new moduli of Remark 5.8 below, Theorem 5.4 applies (with $p = 11$) to 165, 825, 1485, 4125, 6105 and 7425; for the others, such as 105 and 945, no single prime suffices and we rely on computation. We do not have a proof of Conjecture 5.3 in general; the evidence is the following computation.

Theorem 5.6 (computer-assisted)**.** *Every odd non-square $q \le 13105$ is 2-adically obstructed.*

Proof. For primes this is Baker–Birch–Wirsing (or the case $e = 1$ above). For each of the 4938 odd composite non-squares $q \le 13105$ we computed a certificate in the sense of Lemma 4.3 for the shadow system of Theorem 4.5 in the integer form of Remark 4.6: a vector Λ with 2-integral rational entries, one for each unknown $\Phi_d(y)$, which lies in the $\mathbb{Q}$-span of the equations and satisfies $\Lambda(\mathbf{1}) \not\equiv 0 \pmod 2$. If Φ were an odd 2-integral solution, then $0 = \Lambda(\Phi) \equiv \Lambda(\mathbf{1}) \equiv 1 \pmod 2$. The certificates and an independent checker are part of the supplementary material; the checker verifies 2-integrality and oddness directly and membership in the row space by an exact rank computation over $\mathbb{Q}$ (Appendix B). □

The same computation confirms Theorem 5.2 in this range: for each of the 56 odd squares $9 \le q \le 13105$ the shadow system has odd solutions. As cross-checks, the descent was compared with the full system of Theorem 2.2 in all $q-1$ unknowns for 39 moduli up to 2145 (Section 6); and a second program, written independently from the statement of Okada's original equations (it uses neither characters, nor layers, nor the descent), decided the existence of odd Okada vectors for all 332 odd composite $q \le 999$ by two methods (2-adic elimination over $\mathbb{Z}_{(2)}$, and exact kernel saturation): they exist exactly for the 15 odd squares $3^2, 5^2, \dots, 31^2$, in agreement with Theorems 5.2 and 5.6.

Corollary 5.7. *For every odd non-square $q \le 13105$ and every Erdős function f modulo q with $\sum_{n=1}^{q} f(n) = 0$, the number $\sum_{n\ge 1} f(n)/n$ is non-zero, hence transcendental.*

Proof. Theorem 5.6 and Lemma 4.2. □

Remark 5.8. Of the moduli in Corollary 5.7, exactly 74 satisfy none of the sufficient conditions known to us from the literature: q prime [2]; $2\varphi(q)+1 > q$ [10]; $q \equiv 3 \pmod 4$ [9];

$\sum_{d|q,\, d\geq 3} 1/\varphi(d) < 1$ or "least prime factor $\geq$ number of divisors" [4, Prop. 3.1, Cor. 3.3]; and the family $q = pN$ of [3, Cor. 1]. They are all $\equiv 1 \pmod 4$; the first ones are 105, 165, 525, 585, 825, 945, 1365, 1485, 1785, 2145, 2205, 2625, 2805, 2925, 3045, 3465.

5.3. **The least counterexample.**

Theorem 5.9. *Let $2 \leq q \leq 13105$. An Erdős function modulo q with $\sum_{n=1}^{q} f(n) = 0$ and $L(1, f) = 0$ exists if and only if $q = 11025$.*

Proof. If q is even the sum $\sum_{n=1}^{q} f(n)$ has an odd number of terms ± 1 and cannot vanish. If q is an odd non-square, use Theorem 5.6 and Lemma 4.2. If $q = m^2$ is an odd square, then $m \leq 114$, and every odd $m \leq 114$ other than $105 = 3 \cdot 5 \cdot 7$ has at most two distinct prime factors (the next candidate is $3 \cdot 5 \cdot 11 = 165$); hence $\varphi(q)/q = \prod_{p|m}(1 - 1/p) \geq \frac{2}{3} \cdot \frac{4}{5} > \frac{1}{2}$ and Okada's theorem [10] applies. Finally $q = 105^2$ is Theorem 3.1. □

6. How the counterexamples were found, and what failed

6.1. **Symmetric lifting.** By Theorem 4.5(2), an H-invariant Erdős function modulo q is the same thing as a solution of the shadow system with $\Phi_d(y) = |H_d|\, s_d(y)$, $s_d(y) \in \{\pm 1\}$. For a square q the shadow system does have odd solutions (Theorem 5.2), so it is reasonable to look for ± 1 solutions among them. This is a 0/1 feasibility problem with $\prod(1 + e_p 2^{t_p}) - 1$ variables and the integer equations of Remark 4.6; for $q = 105^2$ and $q = 165^2$ there are 224 variables and 19 equations. We solved it with the CP-SAT solver of OR-Tools, version 9.15 [14]. The outcomes, with the time limits used, were as follows.

- $q = 105^2$: a solution was found in about 214 seconds (two earlier runs with limits of 200 and 600 seconds ended without a decision). Minimizing the number of shadow classes on which s_d differs from $\pm\chi_{35}$ on the layers $35 \mid d$ then produced, within 900 seconds, the solution of Table 1, which differs from that pattern only on the layer $d = 105$ and depends on Legendre symbols alone (optimality of this count was not proved).
- $q = 165^2$: a solution was found in about 55 seconds. The subproblems "f depends only on Legendre symbols", "f odd" ($f(-n) = -f(n)$) and "f even" ($f(-n) = f(n)$) are all infeasible (proved by the solver).
- $q = 195^2 = 38025$: the H-invariant problem is infeasible (proved in about one second), so a counterexample modulo 195^2, if any, cannot be H-invariant.
- Legendre-type H-invariant problems (each s_d a function of the Legendre symbols only): solutions were found for 315^2 (in under one second), 735^2 (about 67 seconds) and 495^2 (about 75 seconds), giving Theorem 3.4; the problem is infeasible for 165^2 and 195^2, and was undecided for 525^2 after 600 seconds.
- The squares $225, 441, 2025, 53361, 65025, \ldots$ with $2\varphi(q) + 1 > q$ gave infeasible problems, as they must by Okada's theorem.
- Functions depending only on $\gcd(n, q)$ never give counterexamples for the squares we tried ($15^2, 105^2, 165^2, 195^2, 315^2, 1155^2$): the corresponding sign problems, which only involve the conditions (B′) and (C′), are infeasible.

Each solution was expanded to all $q - 1$ residues and verified exactly as described in Section 3.

6.2. **Relaxations that do not suffice.** The descent of Theorem 4.5 keeps the whole maximal 2-power quotient. Several natural smaller systems are too weak to detect the obstruction for non-squares: keeping only the quadratic characters (in the form valid for relaxations: unknowns are the fibre sums of $b = (1 - f)/2$, and (B′), (C′) become inhomogeneous) misses the obstruction for 116 of the 248 odd composite non-squares $q < 800$, the first being $15, 35, 45$; capping all t_p at 2 still misses 40 of them, the first being $51 = 3 \cdot 17$; keeping only characters supported on one prime misses $15, 35, 45, 51, \ldots$, and on at most two primes misses $105, 195, 231, 255, 945, 1155, 1365$

among the moduli tested; keeping only the odd characters detects exactly the $q \equiv 3 \pmod 4$ among the 19 moduli tested (as Proposition 4.4 predicts), and keeping only the even characters detects nothing. So for $q \equiv 1 \pmod 4$ the obstruction genuinely involves the interaction of odd and even characters, of all 2-power orders, supported on all primes of q at once.

6.3. **Validation of the computations.** The layered system of Theorem 2.2 was validated before use: for $q \le 315$ (fifteen moduli) its rank is $\varphi(q) + \omega(q)$, random rational vectors in its kernel give $|L(1, f)| < 10^{-50}$ in 50-digit arithmetic, and generic vectors do not. The shadow test of Theorem 4.5 agrees with the test on the full system in all $q - 1$ unknowns for the 39 moduli we compared (all 26 odd composite $q \le 105$ and thirteen larger ones up to 2145), and the two forms of the shadow test (unknowns Φ with target "all odd", and unknowns B with inhomogeneous right-hand side) agree for all odd composite $q < 800$. The independent program mentioned after Theorem 5.6 also re-verified the counterexample of Table 1, the first counterexample found modulo 105^2 (it is in the supplement), and the counterexamples modulo 165^2 and 315^2: all of Okada's equations hold exactly, in two different implementations, and $L(1, f)$ vanishes to more than 110 digits by three numerical methods, one of which (exact partial sums up to $40q$ plus a moment expansion of the tail in Hurwitz zeta values) does not use the digamma function; perturbing f at two places makes $L(1, f)$ visibly non-zero.

7. Open problems

(1) *Prove Conjecture 5.3.* It is known for $q \equiv 3 \pmod 4$ and for odd prime powers with odd exponent, and verified for $q \le 13105$. A proof would establish Conjecture 1.1 for all non-square q, together with the transcendence of all the corresponding sums. The relaxation experiments of Section 6 indicate that a proof must use the full 2-power quotient and the interaction of odd and even characters. A natural intermediate statement would be a lifting lemma: if q is obstructed and $\gcd(q, N) = 1$, then qN is obstructed; together with the prime-power case it would imply the conjecture.

(2) *Determine the set $\mathcal{C}$ of counterexample moduli.* We know $105^2, 165^2, 315^2, 495^2, 735^2 \in \mathcal{C}$, that $\mathcal{C}$ is a multiplicative semigroup, that every element of $\mathcal{C}$ is odd, and that $q \notin \mathcal{C}$ whenever $2\varphi(q) + 1 > q$. If Conjecture 5.3 holds, $\mathcal{C}$ is contained in the set of odd squares q with $2\varphi(q) + 1 \le q$. Is $195^2 \in \mathcal{C}$? (No counterexample modulo 195^2 can be invariant under the odd-order units.) Does every odd square q with $2\varphi(q) + 1 \le q$ belong to $\mathcal{C}$?

(3) *Explain the counterexamples.* The function of Table 1 is built from the Jacobi symbol $\left(\frac{x}{35}\right)$ on most layers. Is there a conceptual construction, for instance through Gauss sums or class numbers of the quadratic fields of conductor dividing 105, that produces counterexamples for an infinite family of squares beyond the semigroup generated by 105^2 and 165^2?

(4) *Higher L-values.* The analogous questions for $\sum f(n)/n^k$, $k \ge 2$, are tied to the Chowla–Milnor conjecture [3]; it would be interesting to know whether a 2-adic shadow principle also applies there.

Appendix A. The counterexample modulo 165^2

Table 4: The counterexample modulo $q = 27225 = 3^2 5^2 11^2$. For $n = (q/d)x$ with $\gcd(x, d) = 1$ put $\alpha = \mathrm{ind}_2(x \bmod 3) \bmod 2$, $\beta = \mathrm{ind}_2(x \bmod 5) \bmod 4$, $\gamma = \mathrm{ind}_2(x \bmod 11) \bmod 2$ (only for the primes dividing d; 2 is a primitive root modulo $3, 5, 11$). The string lists $f(n)$ for the tuples (α, β, γ) in lexicographic order (coordinates for primes not dividing d omitted).

d	q/d	coordinates	signs
3	9075	(α)	++
5	5445	(β)	−−−−
9	3025	(α)	++
11	2475	(γ)	−+
15	1815	(α, β)	−−+++−−
25	1089	(β)	+−+−
33	825	(α, γ)	+−+−
45	605	(α, β)	−+−−−+−−
55	495	(β, γ)	+−−++−−+
75	363	(α, β)	++−−+−++
99	275	(α, γ)	−−++
121	225	(γ)	−+
165	165	(α, β, γ)	+−−+−−−−+−−+++++
225	121	(α, β)	−+−++−−+
275	99	(β, γ)	−+−+++−−
363	75	(α, γ)	++−−
495	55	(α, β, γ)	−+−−−+−+−+++−+−+
605	45	(β, γ)	−+−+−+−+
825	33	(α, β, γ)	−+−+−+−+−+−+−+−+
1089	25	(α, γ)	−+−+
1815	15	(α, β, γ)	−+−+−+−+−+−+−+−+
2475	11	(α, β, γ)	−+−+−+−+−+−+−+−+
3025	9	(β, γ)	−+−+−+−+
5445	5	(α, β, γ)	−+−+−+−+−+−+−+−+
9075	3	(α, β, γ)	−+−+−+−+−+−+−+−+
27225	1	(α, β, γ)	+−+−+−+−+−+−+−+−

Appendix B. Certificates and algorithms

Deciding obstruction. For a modulus q we build the integer system of Remark 4.6: rows indexed by the primitive functionals of the $H_m(\Phi)$ for squarefree $m \mid \mathrm{rad}(q)$, $m > 1$, the rows (B′) multiplied by $p - 1$, and the row (C′). We compute a basis of its rational kernel, saturate it at 2 (repeatedly replacing a basis vector by half of an $\mathbb{F}_2$-dependent combination until the reductions modulo 2 are independent), and test whether the all-ones vector lies in the $\mathbb{F}_2$-span of the reductions. By Lemma 4.3 this decides 2-adic obstruction.

Certificates. If q is obstructed, let $k_1, \dots, k_r$ be the saturated kernel basis. Solving the $\mathbb{F}_2$-system $\bar{\Lambda} \cdot k_j = 0$ $(1 \le j \le r)$, $\bar{\Lambda} \cdot \mathbf{1} = 1$ gives $\bar{\Lambda} \in \{0, 1\}^N$; then $\Lambda = \bar{\Lambda} + 2w$, where w is supported on the pivot coordinates of the k_j and solves $w \cdot k_j = -\frac{1}{2}\bar{\Lambda} \cdot k_j$, satisfies $\Lambda \cdot k_j = 0$ for all j (so Λ lies in the row space), has 2-integral entries (the pivot minor is odd), and $\Lambda(\mathbf{1}) \equiv \bar{\Lambda}(\mathbf{1}) = 1$

(mod 2). The supplementary file of certificates contains Λ for each of the 4938 odd composite non-squares $q \le 13105$. The checker rebuilds the integer system from its definition and verifies, for each q: every entry of Λ has odd denominator; $\Lambda(\mathbf{1})$ has odd numerator and denominator; and $\operatorname{rank}\binom{M}{\Lambda} = \operatorname{rank} M$ over $\mathbb{Q}$, computed exactly with FLINT [15]. These checks do not depend on the saturation step.

Verification of the counterexamples. For the two counterexamples we verified, independently of the shadow code: the values are ± 1 off multiples of q and sum to 0; Okada's original equations hold exactly (all $\varphi(q)$ unit equations, with the infinite sums over $M(q)$ evaluated as geometric series, and the $\omega(q)$ prime equations); the coefficients (2.3) vanish for all characters of 2-power order (for 105^2 by hand, Tables 2–3); and $L(1,f) = -\frac{1}{q}\sum_a f(a)\psi(a/q)$ vanishes to 134 significant digits in PARI/GP and to 50 digits in mpmath.

The 74 new moduli of Remark 5.8. 105, 165, 525, 585, 825, 945, 1365, 1485, 1785, 2145, 2205, 2625, 2805, 2925, 3045, 3465, 3705, 3885, 4125, 4305, 4389, 4485, 4641, 4725, 4785, 4845, 5145, 5265, 5313, 5445, 5565, 5865, 5985, 6045, 6105, 6405, 6765, 6825, 7245, 7425, 7605, 7665, 7905, 8085, 8265, 8385, 8505, 8745, 8925, 9009, 9165, 9345, 9405, 9765, 9945, 10005, 10065, 10185, 10545, 10605, 10725, 10965, 11385, 11445, 11505, 11685, 11781, 11865, 11985, 12045, 12285, 12705, 12765, 13065.

Appendix C. Code and data

The supplementary archive contains: the layered criterion and its validation; the shadow construction and the obstruction test in both forms; the certificate generator and checker with the certificate file for $q \le 13105$; the CP-SAT models used to find the counterexamples; the two counterexamples as explicit lists of values $f(1), \dots, f(q-1)$; the independent verification scripts (exact Okada equations, character coefficients, PARI/GP numerics); and logs of all runs quoted in the text.

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